A CASE-STUDY BASED REPORT

ON

Road Segment Risk Classification Using Decision Trees and LightGBM

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Abstract

Road accidents are a major concern due to varying road conditions, traffic patterns, and environmental factors. Identifying high-risk road segments in advance can help authorities improve road safety and reduce accidents. The objective of this project is to classify road segments into different risk categories using machine learning models and determine an effective approach for accurate road risk prediction. The study uses a road segment dataset containing features such as traffic volume, road characteristics, weather conditions, and accident-related information. The dataset is preprocessed by handling missing values, encoding categorical attributes, and scaling the required features. Two machine learning algorithms, Decision Tree and LightGBM (Light Gradient Boosting Machine), are implemented for road segment risk classification. The dataset is divided into training and testing sets. Both models are trained, evaluated, and compared using standard performance metrics such as accuracy, precision, recall, and F1-score to identify the better-performing model. The experimental results demonstrate that both models can classify road segment risks effectively. However, LightGBM achieves higher prediction accuracy and better overall performance compared to the Decision Tree model due to its ensemble learning capability. This project shows that machine learning techniques can effectively classify road segment risks and support proactive road safety planning. The proposed approach can assist transportation authorities in identifying high-risk locations and implementing preventive measures to reduce accidents.

Introduction

Background

Road accidents can occur due to several factors, including traffic conditions, road characteristics, weather conditions, and previous accident patterns. Identifying road segments that have a higher risk of accidents can help authorities take preventive actions and improve overall road safety. Traditional approaches may not effectively capture the complex relationships between different road and environmental factors.

Machine learning provides an effective approach for analyzing road-related data and classifying road segments according to their risk levels. In this project, machine learning techniques are used to classify road segments into different risk categories, focusing on two models: Decision Tree and LightGBM.

Motivation for the Project

Road accidents remain a major public-safety concern, and their likelihood varies from one stretch of road to another with traffic volume, road design, weather and past accident history. Authorities seldom have the resources to treat every segment at once, so knowing in advance which segments are high-risk allows limited effort to be directed where it matters most. Because the relationships among these factors are complex and not easily captured by traditional approaches, a data-driven machine learning approach is a natural fit. The project is motivated by the opportunity to support proactive road safety planning rather than reacting after accidents have already occurred.

Overview of the Work Carried Out

The work carried out in this case study can be summarised as follows:

- A road segment dataset containing traffic volume, road characteristics, weather conditions and accident-related information was used.
- The data was preprocessed by handling missing values, encoding categorical attributes and scaling the required features.
- The dataset was divided into training and testing sets.
- Two models, Decision Tree and LightGBM, were trained for road segment risk classification.
- Both models were evaluated and compared using accuracy, precision, recall and F1-score.

Software requirements used for the implementation:

Component	Details
Programming language	Python 3.x
Development environment	Jupyter Notebook / Google Colab
Operating system	Windows
Pandas	Data handling and preprocessing
NumPy	Numerical computations
Scikit-learn	Decision Tree, data preprocessing and evaluation metrics

LightGBM	Light Gradient Boosting Machine model
Matplotlib	Data visualization
Seaborn	Graphical visualization

Scope and Responsible-Use Statement

The scope of this study is limited to classifying road segments into risk categories from the features available in the chosen dataset, and to comparing a Decision Tree with LightGBM on that task. It does not attempt to predict individual accidents or to assign blame to drivers or authorities.

The predictions are intended as decision support for transportation authorities, for example to prioritise inspections and preventive measures, and should be reviewed by road-safety engineers before action is taken. Because the models learn from historical data, any gaps or bias in that data, such as under-reporting of accidents on some roads, can carry into the predictions and should be kept in mind when interpreting the results.

Problem Statement

Case-Study Scenario

Consider a transportation authority responsible for a large network of road segments. Segments differ in traffic volume, physical characteristics and exposure to weather, and some have a history of accidents. The authority has limited budget and manpower for safety improvements and needs a reliable way to rank segments by risk so that preventive measures can be planned in advance. Using recorded segment data, the task is to build a machine learning model that assigns each segment to a risk category.

Central Case-Study Question

Can road segments be classified accurately into risk categories using traffic, road, weather and accident-related features, and which of the two models, Decision Tree or LightGBM, gives the more effective classification?

Formal Problem Definition

Let $D = \{(x_i, y_i)\}$, $i = 1 \ldots n$, be the road segment dataset, where x_i is the feature vector of the i-th segment (traffic volume, road characteristics, weather conditions and accident-related information) and y_i is its risk category. The goal is to learn a classifier $f: X \rightarrow Y$ that predicts the risk category of an unseen segment with minimum misclassification error. Two classifiers, f_{DT} (Decision Tree) and f_{LGBM} (LightGBM), are trained on a training subset and compared on a held-out testing subset using accuracy, precision, recall and F1-score.

Objectives that Guide the Study

1. To classify road segments into different risk categories using machine learning models.
2. To preprocess the dataset by handling missing values, encoding categorical attributes and scaling the required features.
3. To implement Decision Tree and LightGBM classifiers on the training data.
4. To evaluate and compare both models using accuracy, precision, recall and F1-score.
5. To identify the better-performing approach for accurate road risk prediction and discuss how it can support road safety planning.

Assumptions and Limitations

Assumptions

- The dataset is representative of the road segments for which predictions are needed.
- The risk category labels and the recorded features are reasonably accurate and consistently measured.
- The random split into training and testing sets reflects how the model will meet unseen segments.

Limitations

- Only two algorithms, Decision Tree and LightGBM, are compared; other models are not explored.

- The models can use only the features present in the dataset; factors not recorded in it are not captured.
- Historical accident data may be incomplete or biased, which can affect the predictions.
- Findings may not generalise to other regions or time periods without re-validation.

Dataset and Exploratory Data Analysis

About the Dataset

The study uses a road segment dataset in which each record describes one road segment. The attributes fall into the following groups:

Feature group	Description
Traffic	Traffic volume on the segment
Road characteristics	Physical and structural properties of the road segment
Weather conditions	Environmental conditions affecting the segment
Accident-related information	Information about accidents associated with the segment
Target	Risk category of the road segment

Purpose of EDA

Exploratory Data Analysis (EDA) was carried out to understand the structure and quality of the data before modelling. Its purposes are to identify missing values and inconsistent entries, to find which attributes are categorical and which are numerical, to examine feature distributions and the balance between risk classes, and to study how the features relate to the risk category. These observations guide the preprocessing choices and support the selection of suitable models.

Preparation Steps Actually Performed

1. Handling missing values in the dataset.
2. Encoding categorical attributes into numerical form so that the models can use them.
3. Scaling the required numerical features.
4. Dividing the dataset into training and testing sets for model building and unbiased evaluation.

EDA Exploration - Implementation

The exploration was implemented in Python using Pandas and NumPy for inspecting and summarising the data, and Matplotlib and Seaborn for visualization. The analysis covers dataset shape and data types, summary statistics, missing-value counts, the distribution of risk categories, distributions of the main features, and relationships between the features and the risk category.

Summary of EDA Findings

The exploration showed that the data needed preprocessing before modelling: missing values had to be handled, categorical attributes had to be encoded, and some features had to be scaled. These steps prepared the dataset for the Decision Tree and LightGBM models.

Algorithm Selection and Implementation

Algorithm Selection

Two classifiers were selected so that a simple, interpretable model could be compared with a more powerful ensemble model on the same data.

- Decision Tree: splits the data on feature values to form a tree of if-then rules. It is easy to interpret, handles numerical and categorical data well, and gives a clear baseline.
- LightGBM (Light Gradient Boosting Machine): a gradient boosting framework that builds many trees sequentially, each correcting the errors of the earlier ones. This ensemble learning approach captures complex, non-linear relationships between road and environmental factors and is efficient on large datasets.

Implementation

1. Load the road segment dataset and apply the preprocessing steps described above.
2. Split the data into training and testing sets.
3. Train the Decision Tree classifier (Scikit-learn) on the training set.
4. Train the LightGBM classifier (LightGBM library) on the training set.
5. Predict risk categories for the testing set with both models.
6. Compute accuracy, precision, recall and F1-score for each model and compare them.

Performance Evaluation and Comparison

Accuracy is the fraction of segments classified correctly. Precision is the fraction of segments predicted in a class that truly belong to it, recall is the fraction of segments of a class that are correctly identified, and F1-score is the harmonic mean of precision and recall.

Metric	Decision Tree	LightGBM
Accuracy	[ ]	[ ]
Precision	[ ]	[ ]
Recall	[ ]	[ ]
F1-score	[ ]	[ ]

Both models classify road segment risk effectively. However, LightGBM achieves higher prediction accuracy and better overall performance than the Decision Tree, owing to its ensemble learning capability. The Decision Tree remains useful where easily explained rules are preferred.

Conclusion

This case study applied machine learning to classify road segments into risk categories using traffic volume, road characteristics, weather conditions and accident-related information. After handling missing values, encoding categorical attributes and scaling the required features, Decision Tree and LightGBM models were trained and compared using accuracy, precision, recall and F1-score.

Both models were able to classify road segment risks effectively, but LightGBM delivered higher accuracy and better overall performance because of its ensemble learning capability. The results show that machine learning can support proactive road safety planning, helping transportation authorities identify high-risk locations and put preventive measures in place to reduce accidents.

As future work, the study can be extended with hyperparameter tuning, additional features and other algorithms, and by validating the models on data from other regions.

Signature of the Team members